# Statistics

## Chapter 7 — Instructor Solutions Manual

October 12, 2026 • Total Exercises: 6

**PROBLEM STATEMENT**

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Step 1

DEFINITIONS

Definition **binomial probability**:

$$P(X = k) =_n C_k \cdot \pi^k \cdot (1 - \pi)^{n-k} = \frac{n!}{k!(n-k)!} \cdot \pi^k \cdot (1 - \pi)^{n-k}$$

**Addition rule** for disjoint or mutually exclusive events:

$$P(A \cup B) = P(A \text{ or } B) = P(A) + P(B)$$

**Complement rule**:

$$P(\sim A) = P(\text{not } A) = 1 - P(A)$$

Step 2

SOLUTION

$$n = \text{Sample size} = 15$$
$$\pi = \text{Probability of success} = 80\% = 0.80$$

The number of successes among a fixed number of independent trials with a constant probability of success follows a **binomial distribution**.

(a) Evaluate the definition of binomial probability at $k = 15$:

$$\begin{aligned} P(X = 15) &=_{15} C_{15} \cdot 0.80^{15} \cdot (1 - 0.80)^{15-15} \\ &= \frac{15!}{15!(15-15)!} \cdot 0.80^{15} \cdot 0.20^0 \\ &= 1 \cdot 0.80^{15} \cdot 1 \\ &= 0.80^{15} \\ &\approx 0.0352 \\ &= 3.52\% \end{aligned}$$

Step 3

(b) Evaluate the definition of binomial probability at $k = 9, 10, 11, 12, 13, 14$:

$$
\begin{aligned}
P(X = 9) &=_{15} C_9 \cdot 0.80^9 \cdot (1 - 0.80)^{15-9} \\
&= \frac{15!}{9!(15-9)!} \cdot 0.80^9 \cdot 0.20^6 \\
&= 5005 \cdot 0.80^9 \cdot 0.20^6 \\
&\approx 0.0430 \\
P(X = 10) &=_{15} C_{10} \cdot 0.80^{10} \cdot (1 - 0.80)^{15-10} \\
&= \frac{15!}{10!(15-10)!} \cdot 0.80^{10} \cdot 0.20^5 \\
&= 3003 \cdot 0.80^{10} \cdot 0.20^5 \\
&\approx 0.1032 \\
P(X = 11) &=_{15} C_{11} \cdot 0.80^{11} \cdot (1 - 0.80)^{15-11} \\
&= \frac{15!}{11!(15-11)!} \cdot 0.80^{11} \cdot 0.20^4 \\
&= 1365 \cdot 0.80^{11} \cdot 0.20^4 \\
&\approx 0.1876 \\
P(X = 12) &=_{15} C_{12} \cdot 0.80^{12} \cdot (1 - 0.80)^{15-12} \\
&= \frac{15!}{12!(15-12)!} \cdot 0.80^{12} \cdot 0.20^3 \\
&= 455 \cdot 0.80^{12} \cdot 0.20^3 \\
&\approx 0.2501 \\
P(X = 13) &=_{15} C_{13} \cdot 0.80^{13} \cdot (1 - 0.80)^{15-13} \\
&= \frac{15!}{13!(15-13)!} \cdot 0.80^{13} \cdot 0.20^2 \\
&= 105 \cdot 0.80^{13} \cdot 0.20^2 \\
&\approx 0.2309 \\
P(X = 14) &=_{15} C_{14} \cdot 0.80^{14} \cdot (1 - 0.80)^{15-14} \\
&= \frac{15!}{14!(15-14)!} \cdot 0.80^{14} \cdot 0.20^1 \\
&= 15 \cdot 0.80^{14} \cdot 0.20^1 \\
&\approx 0.1319
\end{aligned}
$$

**Step 4**

Use the addition rule for mutually exclusive events (which is appropriate as it is not possible to obtain two different number of successes in a single sample) and use the result of part (a):

$$\begin{aligned} P(X \geq 9) &= P(X = 9) + P(X = 10) + P(X = 11) + P(X = 12) \\ &\quad + P(X = 13) + P(X = 14) + P(X = 15) \\ &= 0.0430 + 0.1032 + 0.1876 + 0.2501 \\ &\quad + 0.2309 + 0.1319 + 0.0352 \\ &= 0.9819 \end{aligned}$$

Use the complement rule:

$$\begin{aligned} P(X < 9) &= 1 - P(X \geq 9) \\ &= 1 - 0.9819 \\ &= 0.0181 \\ &= 1.81\% \end{aligned}$$

**Step 5**

(c) Use the addition rule for mutually exclusive events (which is appropriate as it is not possible to obtain two different number of successes in a single sample) and use the result of part (a)-(b):

$$\begin{aligned} P(X \geq 12) &= P(X = 12) + P(X = 13) + P(X = 14) + P(X = 15) \\ &= 0.2501 + 0.2309 + 0.1319 + 0.0352 \\ &= 0.6481 \\ &= 64.81\% \end{aligned}$$

**Result**

(a) $0.0352 = 3.52\%$

(b) $0.0181 = 1.81\%$

(c) $0.6481 = 64.81\%$

**PROBLEM STATEMENT**

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Step 1

DEFINITIONS

Formula **Poisson probability**:

$$P(X = k) = \frac{\mu^k e^{-\mu}}{k!}$$

**Complement rule**:

$$P(\sim A) = P(\text{not } A) = 1 - P(A)$$

**Addition rule** for disjoint or mutually exclusive events:

$$P(A \cup B) = P(A \text{ or } B) = P(A) + P(B)$$

Step 2

SOLUTION

$$n = \text{Sample size} = 400$$
$$\pi = \text{Probability of success} = 0.005$$

(a) The mean is the product of the sample size and the probability of success:

$$\mu = n\pi = 400(0.005) = 2$$

Evaluate the probability of Poisson probability at $x = 0$:

$$\begin{aligned} P(X = 0) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{2^0 e^{-2}}{0!} \\ &= \frac{1e^{-2}}{1} \\ &= e^{-2} \\ &\approx 0.1353 \\ &= 13.53\% \end{aligned}$$

Step 3

(b) The mean is the product of the sample size and the probability of success:

$$\mu = n\pi = 400(0.005) = 2$$

Thus 2 of the 400 loans are expected to default, which implies that $400 - 2 = 398$ loans are not expected to default.

Step 4

(c) The mean is the product of the sample size and the probability of success:

$$\mu = n\pi = 400(0.005) = 2$$

Evaluate the probability of Poisson probability at $x = 1, 2$:

$$\begin{aligned} P(X = 1) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{2^1 e^{-2}}{1!} \\ &= \frac{2e^{-2}}{1} \\ &= 2e^{-2} \\ &\approx 0.2707 \\ P(X = 2) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{2^2 e^{-2}}{2!} \\ &= \frac{4e^{-2}}{2} \\ &= 2e^{-2} \\ &\approx 0.2707 \end{aligned}$$

Step 5

Use the addition rule for mutually exclusive events (which is appropriate as it is not possible to obtain two different number of successes for the same sample) and the results of part (a):

$$\begin{aligned} P(X < 3) &= P(X = 0) + P(X = 1) + P(X = 2) \\ &= 0.1353 + 0.2707 + 0.2707 \\ &= 0.6767 \end{aligned}$$

Use the complement rule:

$$\begin{aligned} P(X \geq 3) &= 1 - P(X < 3) \\ &= 1 - 0.6767 \\ &= 0.3233 \\ &= 32.33\% \end{aligned}$$

🎯 Result

(a) $e^{-2} \approx 0.1353 = 13.53\%$

(b) 398 loans

(c) $0.3233 = 32.33\%$

**PROBLEM STATEMENT**

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Step 1

Given:

$$\mu = \text{Mean} = 10000$$
$$\sigma = \text{Standard deviation} = 2000$$

(a) The z-score is the value decreased by the mean, divided by the standard deviation.

$$z = \frac{x - \mu}{\sigma} = \frac{13500 - 10000}{2000} = 1.75$$

Determine the corresponding probability using the normal probability table in the appendix. $P(0 < Z < 1.75)$ is given in the row starting with $1.7$ and in the column starting with .05 of the standard normal probability table in the appendix.

$$\begin{aligned} P(X > 13500) &= P(Z > 1.75) \\ &= 1 - P(Z < 1.75) \\ &= 1 - P(Z < 0) - P(0 < Z < 1.75) \\ &= 1 - 0.5000 - 0.4599 \\ &= 0.0401 \\ &= 4.01\% \end{aligned}$$

Step 2

(b) The z-score is the value decreased by the mean, divided by the standard deviation.

$$z = \frac{x - \mu}{\sigma} = \frac{8000 - 10000}{2000} = -1$$
$$z = \frac{x - \mu}{\sigma} = \frac{11500 - 10000}{2000} = 0.75$$

Determine the corresponding probability using the normal probability table in the appendix. $P(0 < Z < 0.75)$ is given in the row starting with $0.7$ and in the column starting with .05 of the standard normal probability table in the appendix. $P(0 < Z < 1)$ is given in the row starting with $1$ and in the column starting with .01 of the standard normal probability table in the appendix.

$$\begin{aligned}
&P(8000 < X < 11500) \\
&= P(-1 < Z < 0.75) \\
&= P(Z < 0.75) - P(Z < -1) \\
&= P(Z < 0.75) - P(Z > 1) \\
&= P(Z < 0.75) - (1 - P(Z < 1)) \\
&= P(Z < 0) + P(0 < Z < 0.75) - (1 - P(Z < 0) - P(0 < Z < 1)) \\
&= 0.5000 + 0.2734 - (1 - 0.5000 - 0.3413) \\
&= 0.7734 - 0.1587 \\
&= 0.6147 \\
&= 61.47\%
\end{aligned}$$

Step 3

(c) Let us determine the z-score that corresponds with a probability of $100\% - 10\% - 50\% = 40\%$ in the normal probability table of the appendix (the z-score corresponding with the bottom 10% is then the negative of this z-score). We note that the closest probability is 0.3997 which lies in the row $1.2$ and in the column .08 of the normal probability table and thus the corresponding z-score is then $1.2 + .08 = 1.28$.

$$z = 1.28$$

The z-score is the value decreased by the mean, divided by the standard deviation.

$$z = \frac{x - \mu}{\sigma} = \frac{x - 10000}{2000}$$

The two found expressions of the z-score then have to be equal:

$$\frac{x - 10000}{2000} = 1.28$$

Multiply each side by 2000:

$$x - 10000 = 1.28(2000)$$

Add 10000 to each side:

$$x = 10000 - 1.28(2000)$$

Evaluate:

$$x = 7440$$

Thus 10% of the games have an attendance of 7440 or less.

Result

(a) $0.0401 = 4.01\%$

(b) $0.6147 = 61.47\%$

(c) 7440 or less

**PROBLEM STATEMENT**

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**Step 1**

Given:

| $x$ | 0 | 40 | 300 |
|---|---|---|---|
| $P(X = x)$ | 0.98 | 0.016 | 0.004 |

(a) The expected value (or mean) is the sum of the product of each possibility $x$ with its probability $P(x)$:

$$\begin{aligned}\mu &= \sum xP(x) \\ &= 0 \times 0.98 + 40 \times 0.016 + 300 \times 0.004 \\ &= 1.84\end{aligned}$$

Thus the expected amount of loss is 1.84 millions.

**Step 2**

(b) By part (a), we know that the expected loss is 1.84 million.

The only ways to lose less than 1.8 million is when we makes a loss of 0 million, which has a probability of 0.98.

$$P(X < 1.8) = P(X = 0) = 0.98 = 98\%$$

Step 3

(c) Daniel-James suffers a loss when he loses either 40 million or 300 million.

$$P(X < 1.8) = P(X = 0) = 0.98 = 98\%$$

Use the **Addition rule** for disjoint or mutually exclusive events: $P(A \cup B) = P(A \text{ or } B) = P(A) + P(B)$

$$\begin{aligned} P(\text{loss}) &= P(X = 40) + P(X = 300) \\ &= 0.016 + 0.004 \\ &= 0.020 \end{aligned}$$

Use the Definition of **Conditional probability**: $P(B|A) = \dfrac{P(A \cap B)}{P(A)} = \dfrac{P(A \text{ and } B)}{P(A)}$

$$\begin{aligned} P(\$300 \text{ million}|\text{loss}) &= \frac{P(\$300 \text{ million and loss})}{P(\text{loss})} \\ &= \frac{P(\$300 \text{ million})}{P(\text{loss})} \\ &= \frac{0.004}{0.020} \\ &= \frac{4}{20} \\ &= \frac{1}{5} \\ &= 0.2 \\ &= 20\% \end{aligned}$$

Step 4

(d) The premium of $\$2.0$ million exceeds the expected loss of $\$1.8$ million and thus the premium is fair.

However, this premium does not cover the risk, because if Daniel-James makes a loss, then this loss will far exceed $\$2.0$ million.

🎯 **Result**

(a) 1.84 millions

(b) $0.98 = 98\%$

(c) $\frac{1}{5} = 0.2 = 20\%$

(d) Fair, Not covered

**PROBLEM STATEMENT**

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Step 1

Given:

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Percent | 40% | 30% | 15% | 10% | 5% |
| Probability | 0.40 | 0.30 | 0.15 | 0.10 | 0.05 |

(a) The expected value (or mean) is the sum of the product of each possibility $x$ with its probability $P(x)$:

$$\begin{aligned}\mu &= \sum xP(x) \\ &= 0 \times 0.40 + 1 \times 0.30 + 2 \times 0.15 + 3 \times 0.10 + 4 \times 0.05 \\ &= 1.1\end{aligned}$$

The variance is the expected value of the squared deviation from the mean:

$$\begin{aligned}\sigma^2 &= \sum (x-\mu)^2 P(x) \\ &= (0-1.1)^2 \times 0.40 + (1-1.1)^2 \times 0.30 + (2-1.1)^2 \times 0.15 \\ &+ (3-1.1)^2 \times 0.10 + (4-1.1)^2 \times 0.05 \\ &= 1.39\end{aligned}$$

The standard deviation is the square root of the variance:

$$\sigma = \sqrt{\sigma^2} = \sqrt{1.39} \approx 1.1790$$

Step 2

(b) By part (a), the expected number of school-age children per family in Whitehall Estates is 1.1 child, as the mean is 1.1.

$$\text{Number of families} = 500$$

The expected number of children is then the product of the expected number of children per family and the number of families:

$$\begin{aligned}&\text{Expected number of children}\\&= \text{Expected number of children per family}\\&\times \text{Number of families}\\&= 1.1 \times 500\\&= 500\end{aligned}$$

Step 3

(c) Since we are interested in the expected number of children among families with children, we exclude the case of 0 children.

However, we then also need to adjust the probabilities by dividing the current probability by $0.30 + 0.15 + 0.10 + 0.05 = 0.60$ (total probability of a family having a child.

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| Probability | $\frac{0.30}{0.60} = \frac{1}{2}$ | $\frac{0.15}{0.60} = \frac{1}{4}$ | $\frac{0.10}{0.60} = \frac{1}{6}$ | $\frac{0.05}{0.60} = \frac{1}{12}$ |

The expected value (or mean) is the sum of the product of each possibility $x$ with its probability $P(x)$:

$$\begin{aligned}\mu &= \sum xP(x)\\&= 1 \times \frac{1}{2} + 2 \times \frac{1}{4} + 3 \times \frac{1}{6} + 4 \times \frac{1}{12}\\&= \frac{11}{6}\\&\approx 1.8333\end{aligned}$$

🎯 **Result**

(a) Mean 1.1 and Standard deviation 1.1790

(b) 500 children

(c) $\frac{11}{6} \approx 1.8333$

**PROBLEM STATEMENT**

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Step 1

Given:

| | Democrats | Republicans | Total |
|---|---|---|---|
| House | 236 | 199 | 435 |
| Senate | 48 | 52 | 100 |
| Total | 284 | 251 | 535 |

(a) The table describes 535 people in total, because 535 is mentioned in the bottom right corner of the given table.

251 of the 535 people are Republican, because 251 is mentioned in the row "Total" and in the column "Republicans."
The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{Republican}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{251}{535} \approx 0.4692 = 46.92\%$$

Step 2

(b) The table describes 535 people in total, because 535 is mentioned in the bottom right corner of the given table.

435 of the 535 people are members of the House of Representatives, because 435 is mentioned in the row "House" and in the column "Total."

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{House}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{435}{535}$$

199 of the 535 people are republicans and members of the House of Representatives, because 199 is mentioned in the row "House" and in the column "Republicans."

$$P(\text{House and Republican}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{199}{535}$$

Use the Definition of **Conditional probability**: $P(B|A) = \dfrac{P(A \cap B)}{P(A)} = \dfrac{P(A \text{ and } B)}{P(A)}$

$$\begin{aligned} P(\text{Republican}|\text{House}) &= \frac{P(\text{House and Republican})}{P(\text{House})} \\ &= \frac{199/535}{435/535} \\ &= \frac{199}{435} \\ &\approx 0.4575 \\ &= 45.75\% \end{aligned}$$

Step 3

(c) The table describes 535 people in total, because 535 is mentioned in the bottom right corner of the given table.

435 of the 535 people are members of the House of Representatives, because 435 is mentioned in the row "House" and in the column "Total."

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{House}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{435}{535}$$

284 of the 535 people are Democrats and members of the House of Representatives, because 284 is mentioned in the row "Total" and in the column "Democrats."

$$P(\text{Democrat}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{284}{535}$$

236 of the 535 people are Democrats and members of the House of Representatives, because 236 is mentioned in the row "House" and in the column "Democrats."

$$P(\text{House and Democrat}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{236}{535}$$

Use the **General addition rule** for any two events: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

$$\begin{aligned}
&P(\text{House and Democrat}) \\
&= P(\text{House}) + P(\text{Democrat}) - P(\text{House and Democrat}) \\
&= \frac{435}{535} + \frac{284}{535} - \frac{236}{535} \\
&= \frac{483}{535} \\
&\approx 0.9028 \\
&= 90.28\%
\end{aligned}$$

Result

(a) $\frac{251}{535} \approx 0.4692 = 46.92\%$

(b) $\frac{199}{435} \approx 0.4575 = 45.75\%$

(c) $\frac{483}{535} \approx 0.9028 = 90.28\%$